
PROTOCOL

1. For each reaction you set up, **top-label a PCR tube** with the name(s) indicated on the labsheet.
2. Retrieve your **oligos** (10 μ M) and **template** from the freezer and thaw them at room temperature.
3. Set up the reaction, in order:
 - 32 ddH₂O
 - 10 5 $\times$ PrimeSTAR GXL buffer
 - 4 dNTP mix (2.5 mM each)
 - 1 primer 1 (10 μ M)
 - 1 primer 2 (10 μ M)
 - 1 template
 - 1 PrimeSTAR GXL polymerase

50 μ L total
 - More than about **4 reactions**: make a master mix of everything except template — **49 μ L per tube** plus ~10% overage — then add template to each tube separately.
 - Template is usually a **20 $\times$ dilution** of a miniprep.
4. Retrieve the **enzyme cooler**, and add **1 μ L polymerase** to each sample, last.
5. **Cap, slam on the bench** to mix, **quick spin**.
6. Run the program from your labsheet — it depends on insert length and annealing temperature.